

SUPSI

KrostChain - Document Certification

Platform on Blockchain

Students

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Degree Course	Modules	Accademic Year
Bachelor Ingegneria Informatica	M-I6190Z - Blockchain Engineering Fondamenti e Applicazioni 24-25	2024 - 2025

Stakeholders

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Date

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Introduction

Document Certification on Blockchain

- Upload and register documents
- View the full certification history
- Certify or Revoke (based on user role)



Context and Motivation

Why Blockchain for Document Certification?

- Paper can be forged
- No shared, transparent control
- No proof of integrity or timing
- Trust depends on central authorities



Problem

The Problem We Address

- Lack of verifiable authenticity
- No shared certification history
- High risk of undetected tampering
- No role separation or decentralization



State of the Art

Existing Solutions

Centralized



Trust relies on a central authority

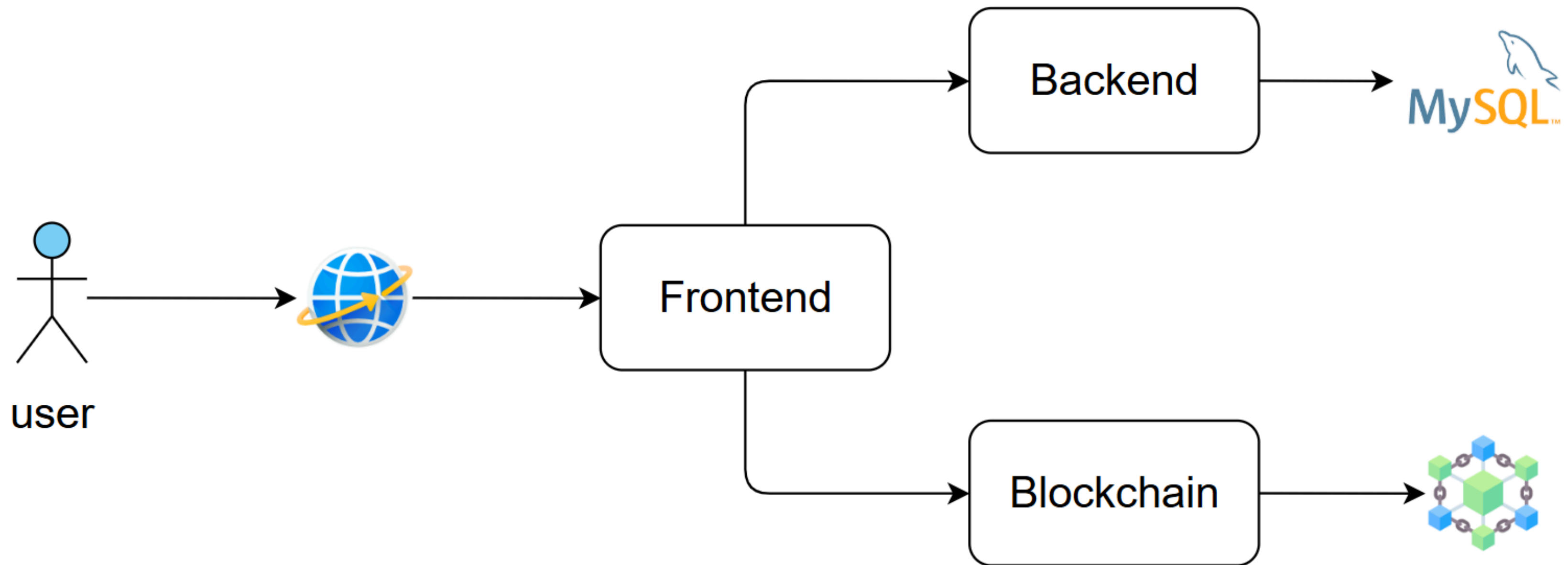
Decentralized



For advanced users, requires library usage

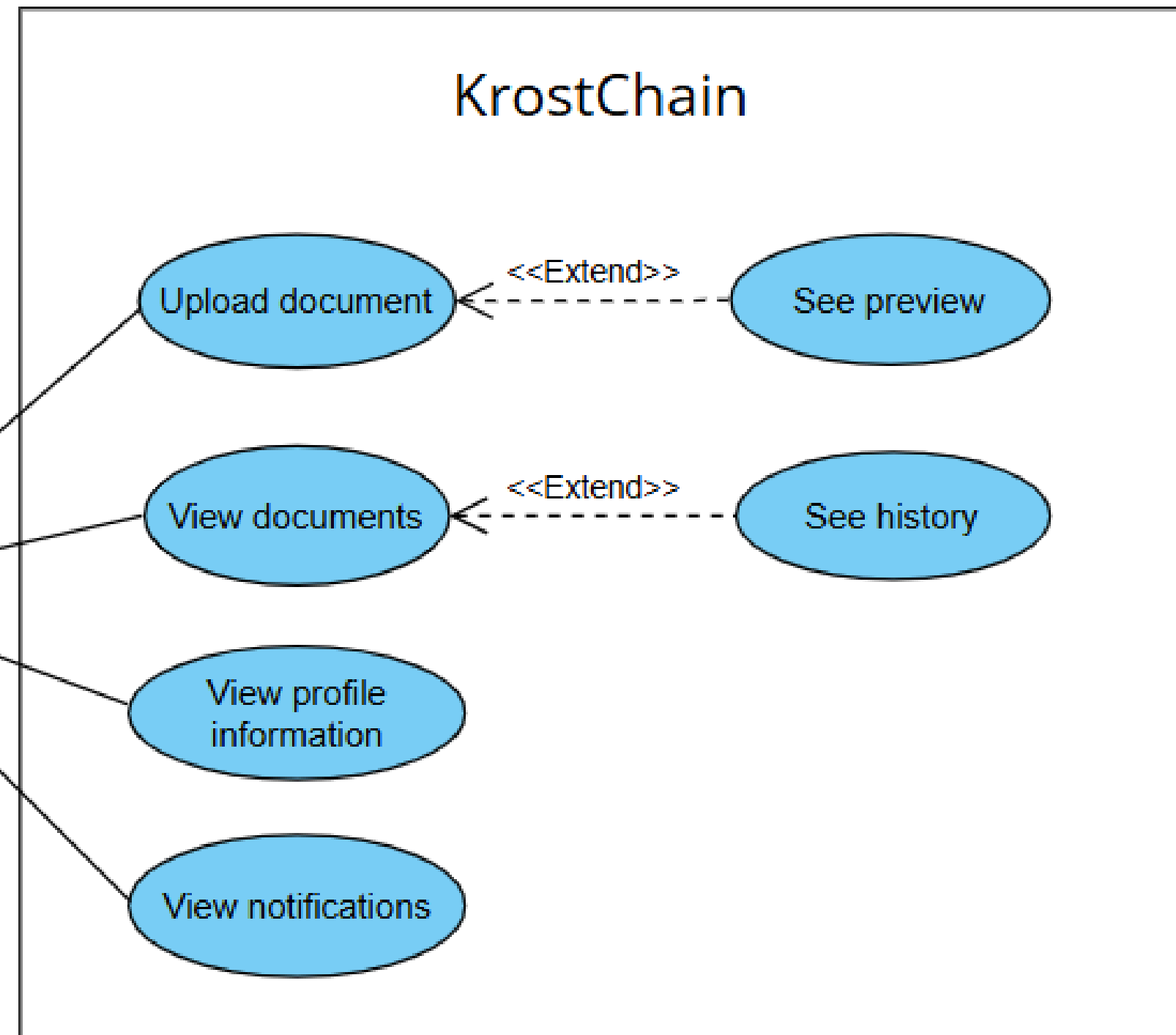
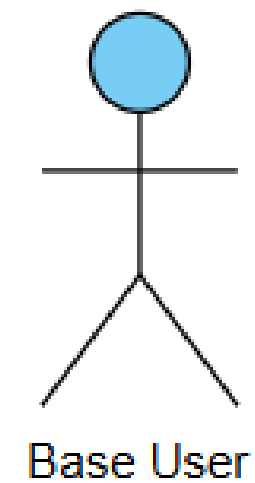
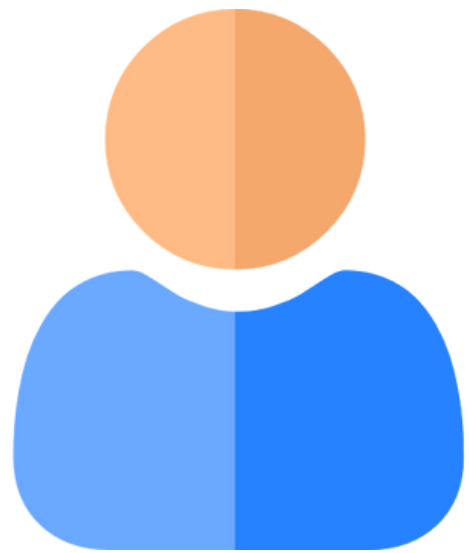
Approach

Hybrid solution



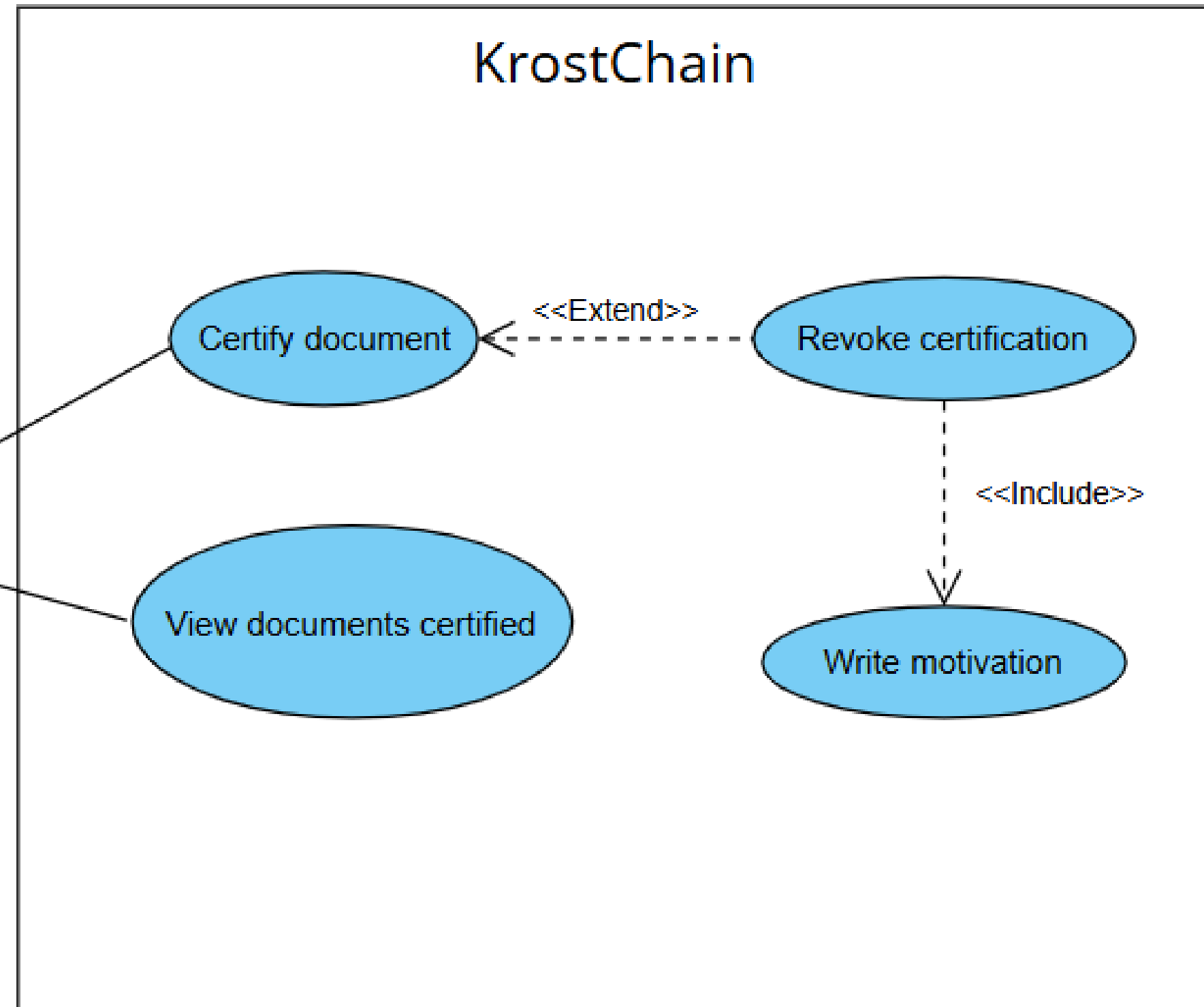
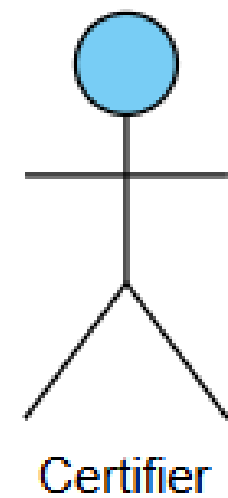
Approach

Use Case Diagram: Base User



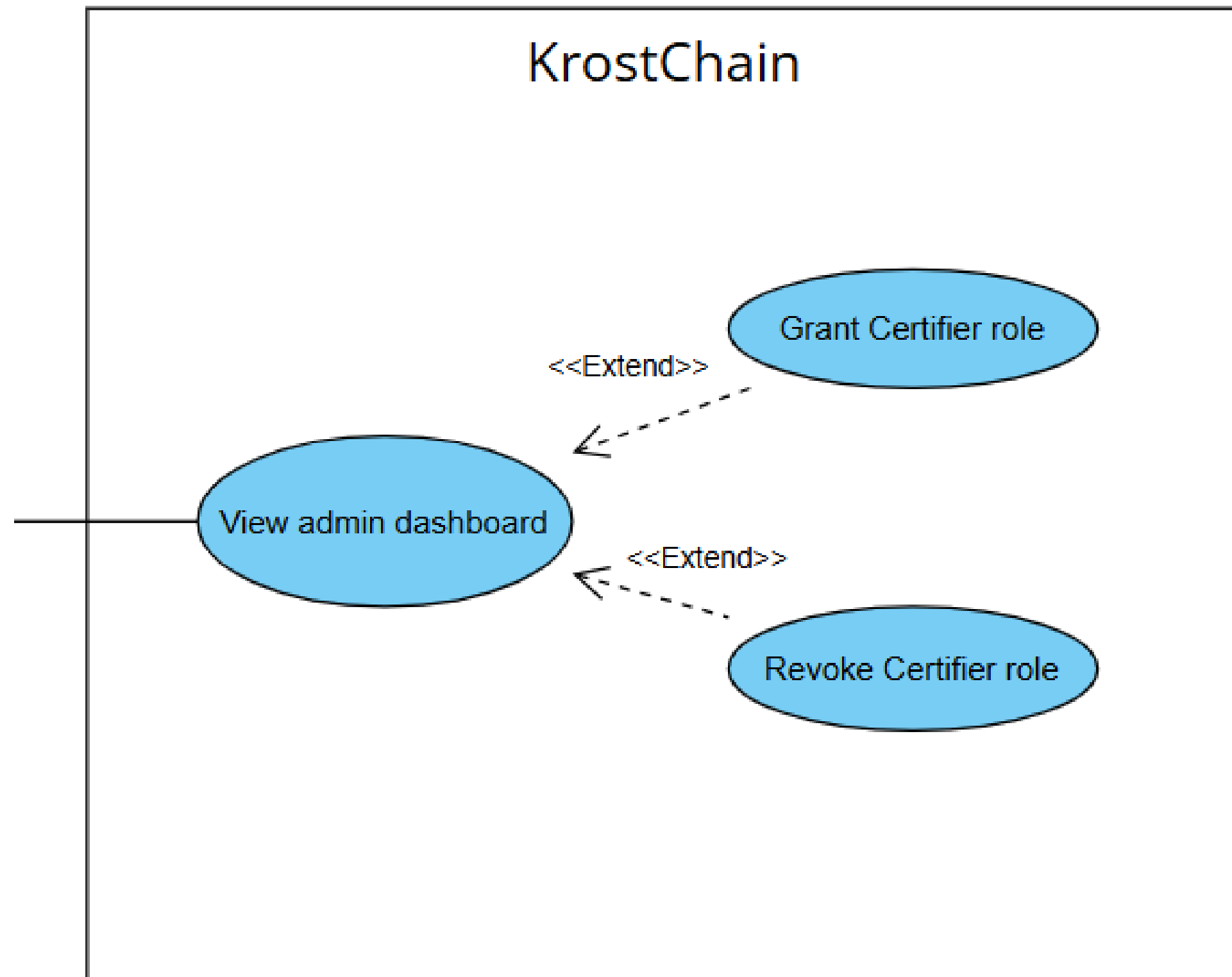
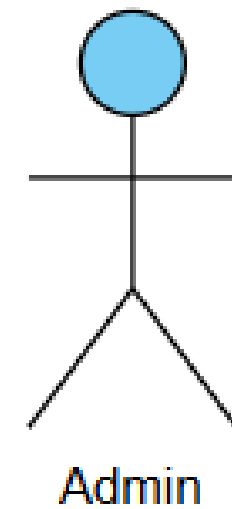
Approach

Use Case Diagram: Certifier



Approach

Use Case Diagram: Admin



Approach

OpenZeppelin



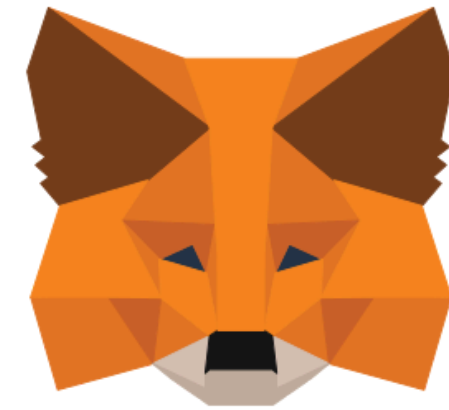
Handle authorizations, with full on-chain control



Safe and modular contract upgrades

Approach

Login with Metamask



METAMASK



Users authenticate via their Ethereum wallet (no passwords)



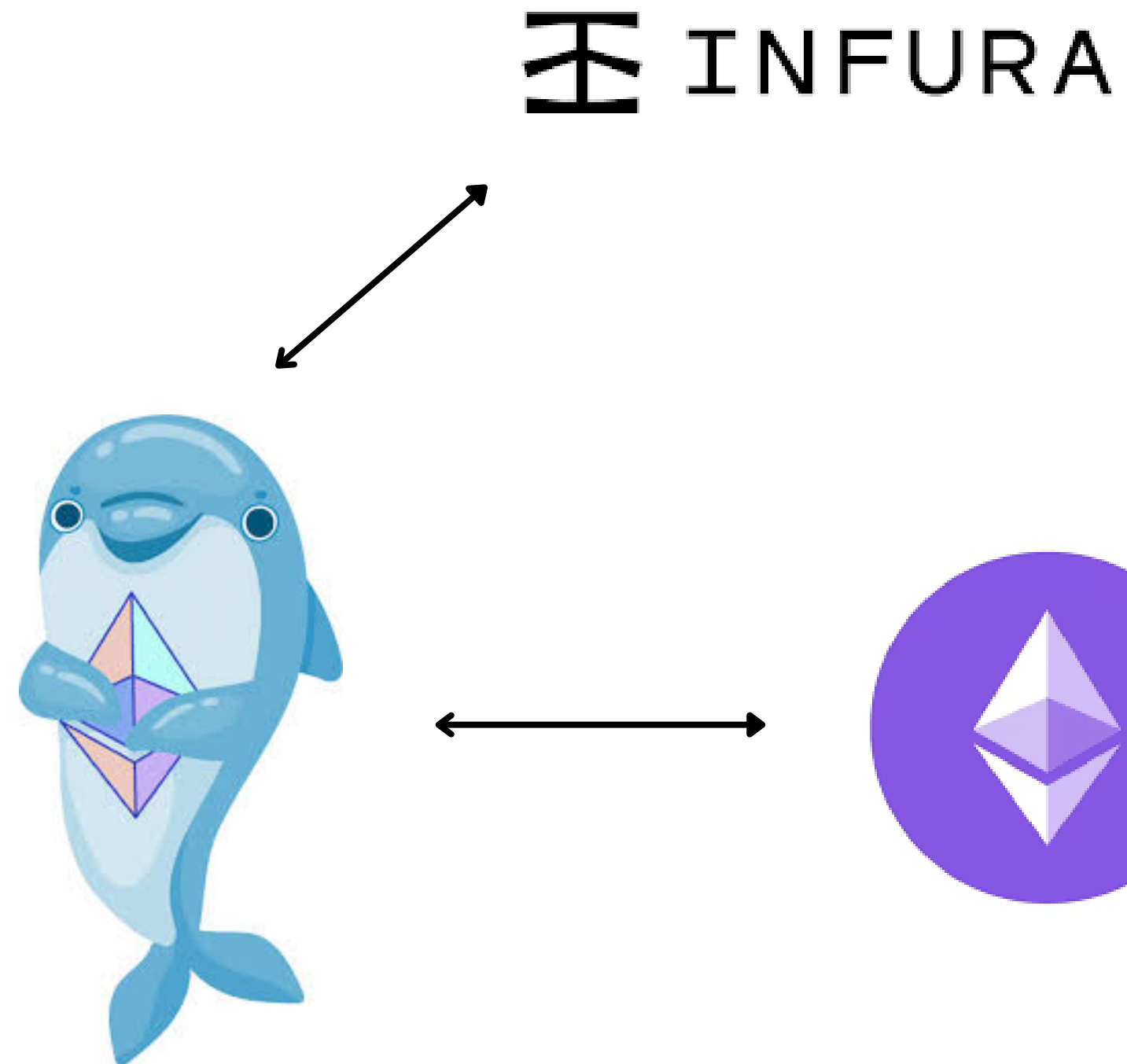
Transactions (certify/revoke) are signed directly in Metamask

Approach

Sepolia Testnet

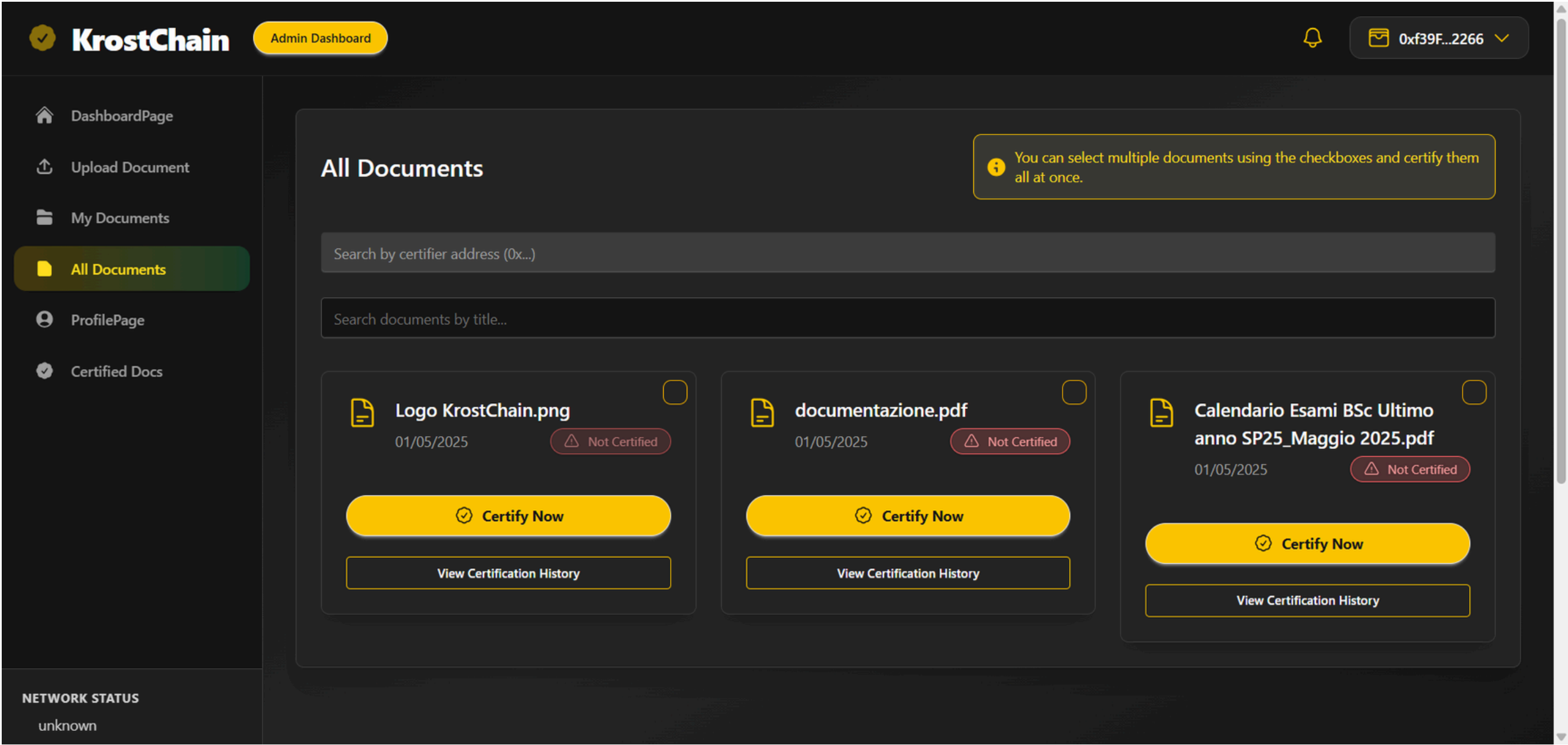
Free test ETH via faucets

Identical environment to mainnet



Results

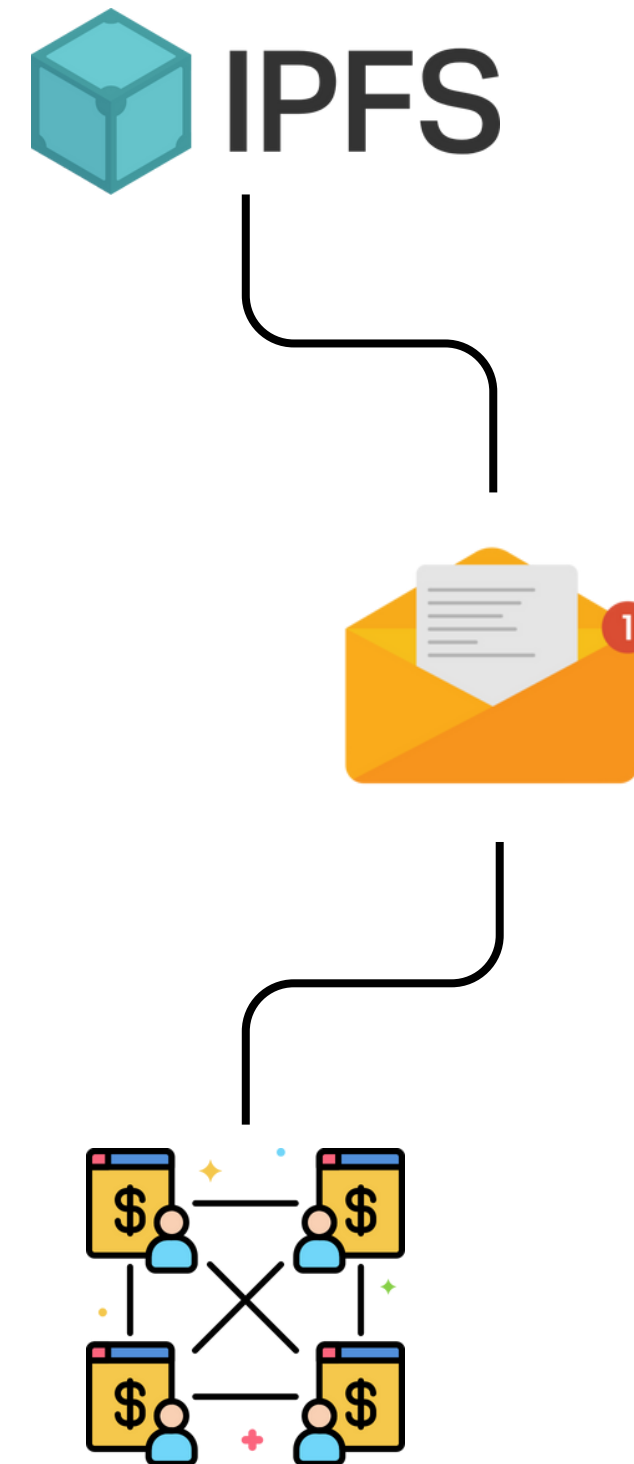
Static demo



Results

Future Focus Areas

- IPFS Integration
- Email Notifications System
- Decentralized Role Governance



THANKS FOR YOUR ATTENTION

